

Preliminary

Project Documentation

C64 Expansion Board Extension Cartridge Side

Project number: 227

Revision: 0

Date: 25.05.2026

C64 Expansion Port Extension Cartridge Side Rev. 0

Module Description

Introduction

The purpose of the C64 Expansion Port Extension is to place an Expansion Port connector at a desired location and connect the actual C64 Expansion Port via a ribbon cable. The maximum length of said ribbon cable is not yet determined. This board is providing the cartridge connector.

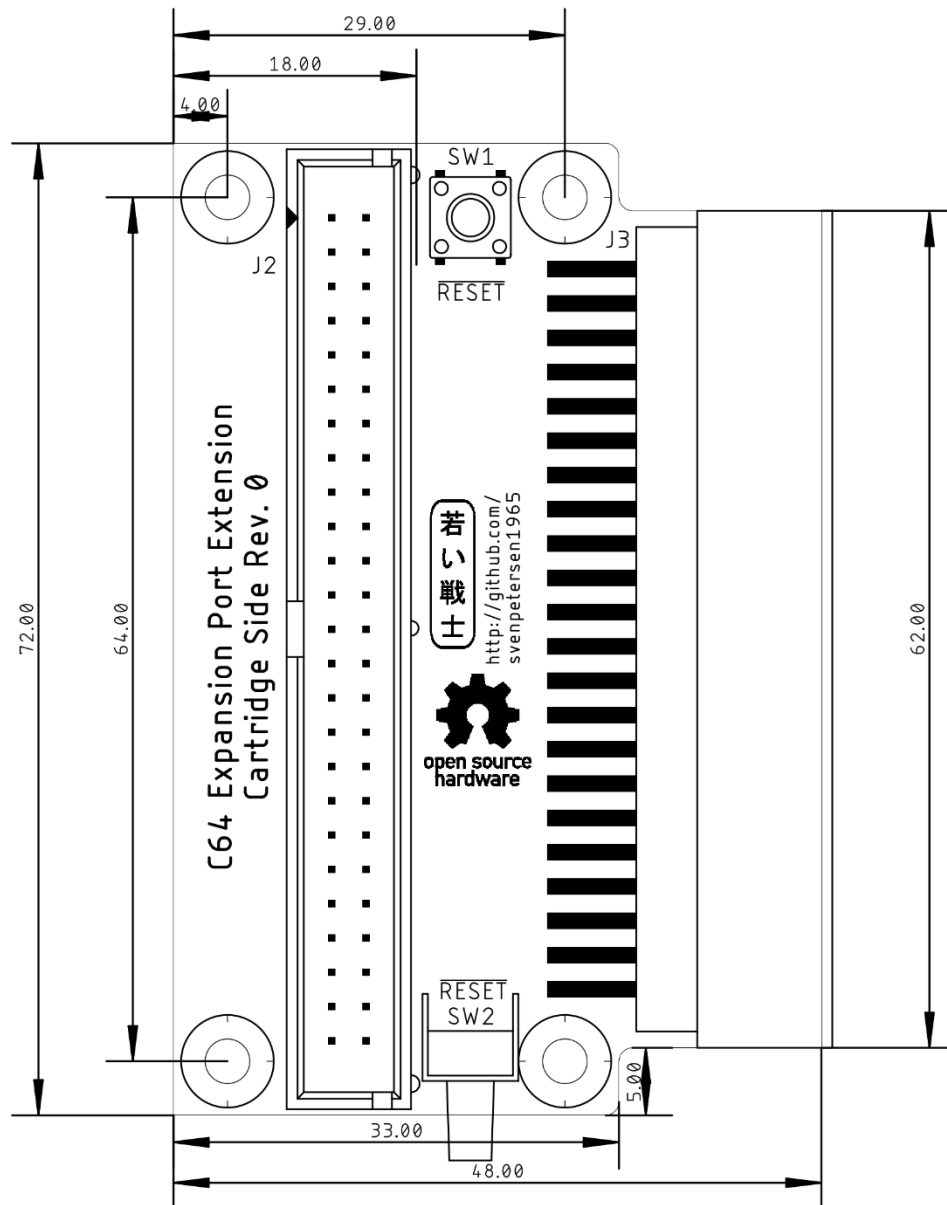


Figure 1: Dimension of the cartridge side PCB

Connector Pinouts

J2 – Break out connector

2x25 box header, 2.54mm pitch

C64ExpExtC_ModD.docx

Drafted by Sven Petersen

Pin	Signal	Pin	Signal
1	GND	2	GND
3	+5V	4	GND
5	+5V	6	$\overline{\text{ROMH}}$
7	+5V	8	$\overline{\text{RESET}}$
9	$\overline{\text{IRQ}}$	10	$\overline{\text{NMI}}$
11	R / $\overline{\text{W}}$	12	PHI2
13	DOT CLOCK	14	A15
15	$\overline{\text{I/O1}}$	16	A14
17	$\overline{\text{GAME}}$	18	A13
19	$\overline{\text{EXROM}}$	20	A12
21	$\overline{\text{I/O2}}$	22	A11
23	$\overline{\text{ROML}}$	24	A10
25	BA	26	A9
27	$\overline{\text{DMA}}$	28	A8
29	D7	30	A7
31	D6	32	A6
33	D5	34	A5
35	D4	36	A4
37	D3	38	A3
39	D2	40	A2
41	D1	42	A1
43	D0	44	A0
45	GND	46	GND
47	GND	48	GND
49	GND	50	GND

J3 – Cartridge Connector

Card Edge Connector 22x2P 2.54mm (0.1") Right Angle

Pin (TOP)	Signal	Pin	Signal (BOT)
1	GND	A	GND
2	+5V	B	$\overline{\text{ROMH}}$
3	+5V	C	$\overline{\text{RESET}}$
4	$\overline{\text{IRQ}}$	D	$\overline{\text{NMI}}$
5	R / $\overline{\text{W}}$	E	PHI2
6	DOT CLOCK	F	A15
7	$\overline{\text{I/O1}}$	H	A14
8	$\overline{\text{GAME}}$	J	A13
9	$\overline{\text{EXROM}}$	K	A12

Pin (TOP)	Signal	Pin	Signal (BOT)
10	$\overline{\text{I/O2}}$	L	A11
11	$\overline{\text{ROML}}$	M	A10
12	BA	N	A9
13	$\overline{\text{DMA}}$	P	A8
14	D7	R	A7
15	D6	S	A6
16	D5	T	A5
17	D4	U	A4
18	D3	V	A3
19	D2	W	A2
20	D1	X	A1
21	D0	Y	A0
22	GND	Z	GND

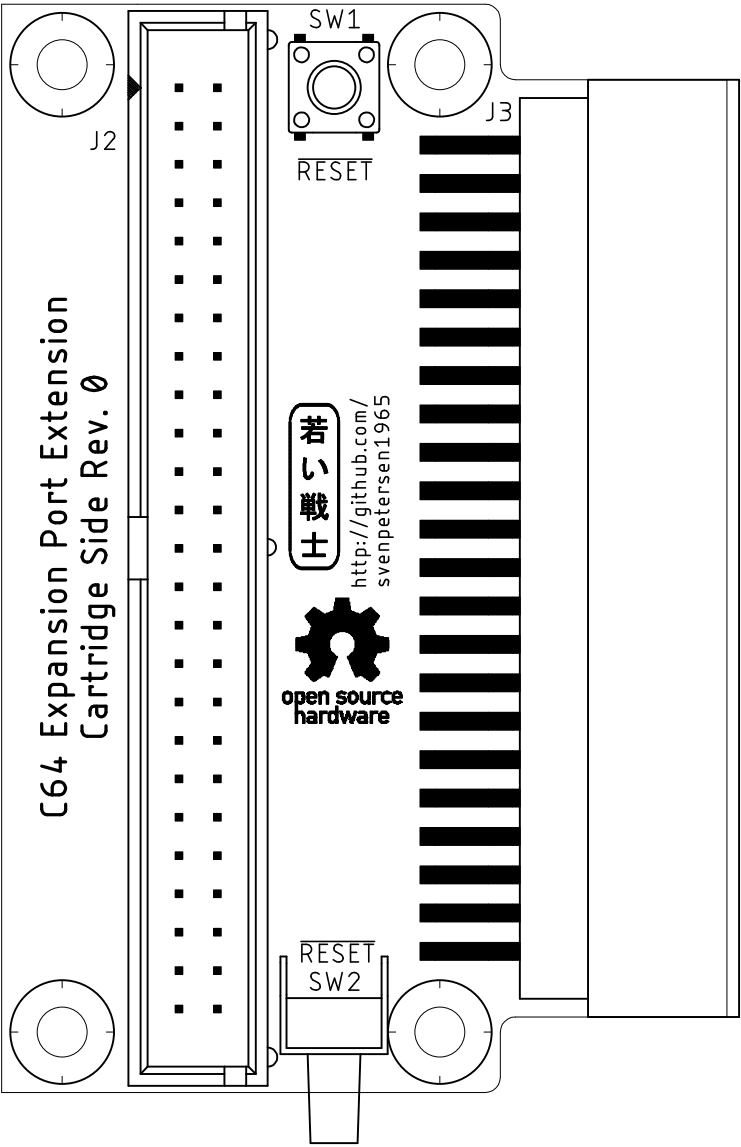
Revision History

Rev. 0

Prototype

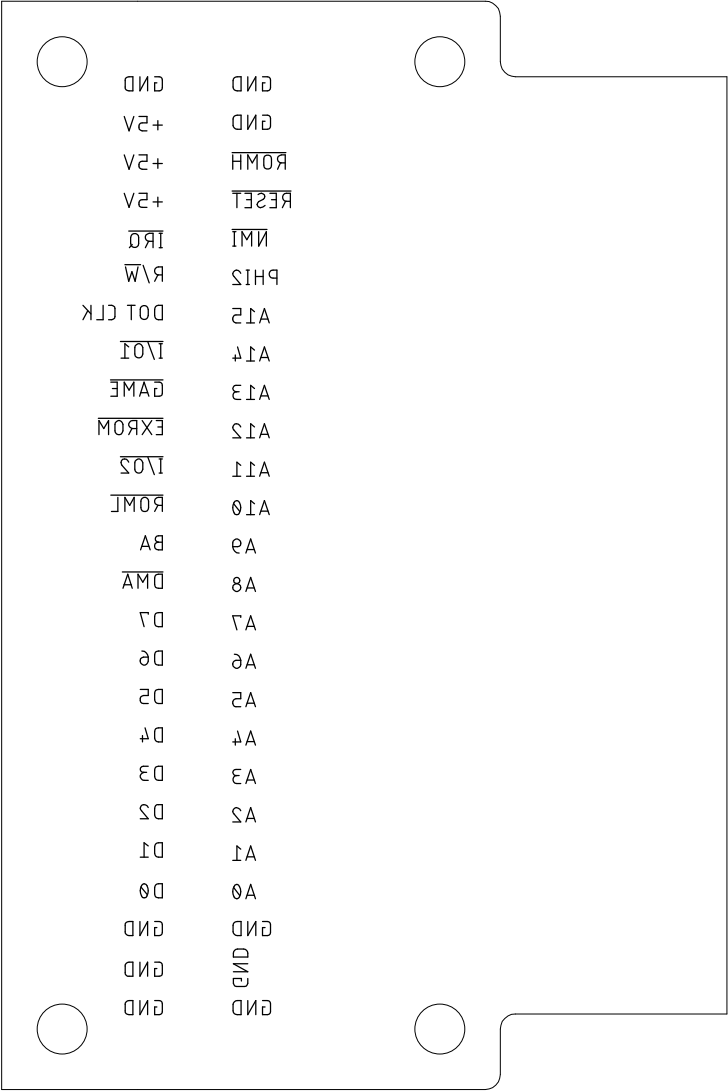
Sven Petersen 2026	Doc.-No.: 227-2-01-00	
	Cu: 35μm	Cu-Layers: 2
C64_Exp_cartridge		
22.05.2026 15:39		Rev.: 0
placement component side		

Preliminary



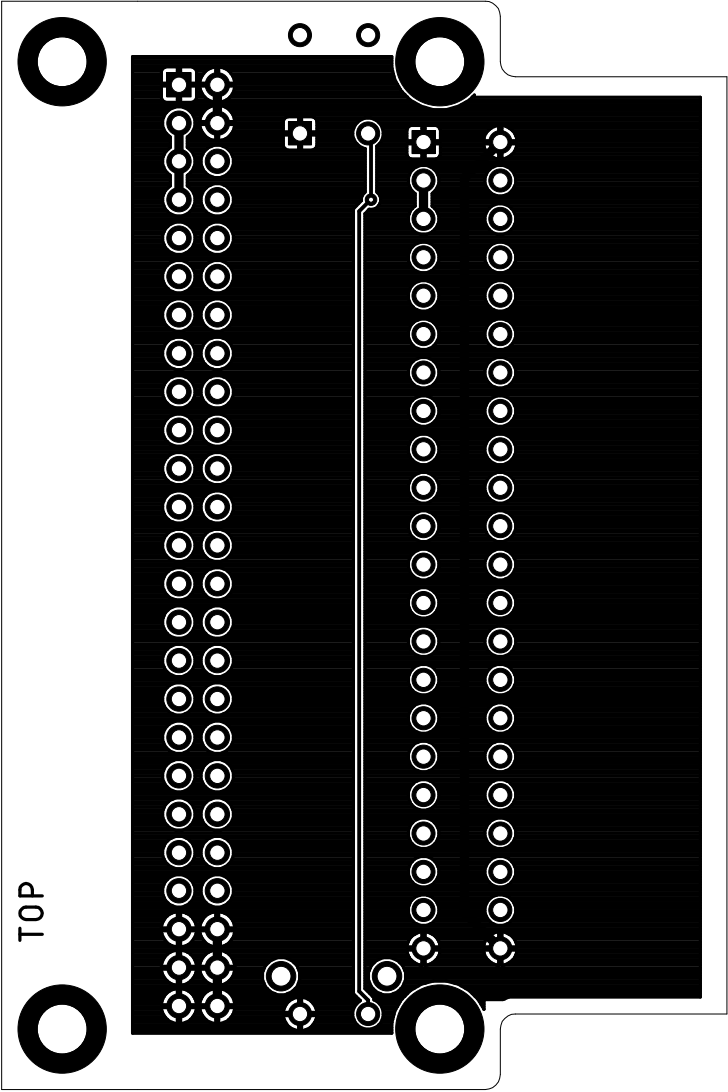
Sven Petersen 2026	Doc.-No.: 227-2-01-00	
	Cu: 35µm	Cu-Layers: 2
C64_Exp_cartridge		
22.05.2026 15:39		Rev.: 0
placement solder tabs		

Preliminary



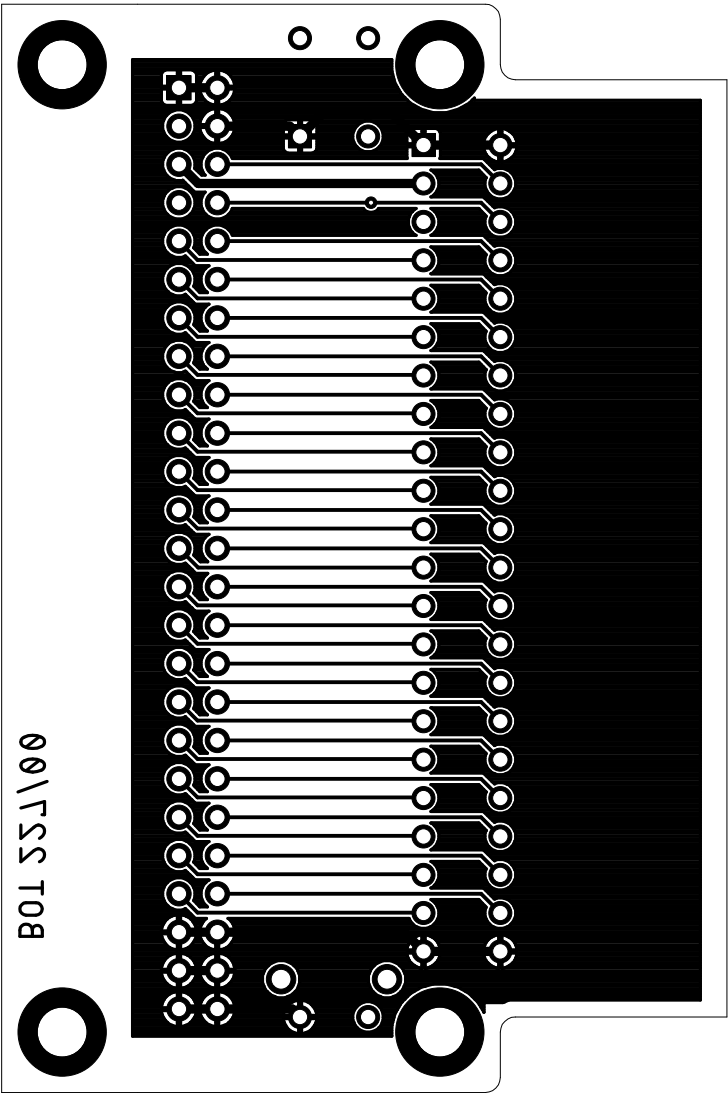
Sven Petersen 2026	Doc.-No.: 227-2-01-00	
	Cu: 35µm	Cu-Layers: 2
C64_Exp_cartridge		
22.05.2026 15:39		Rev.: 0
top		

Preliminary



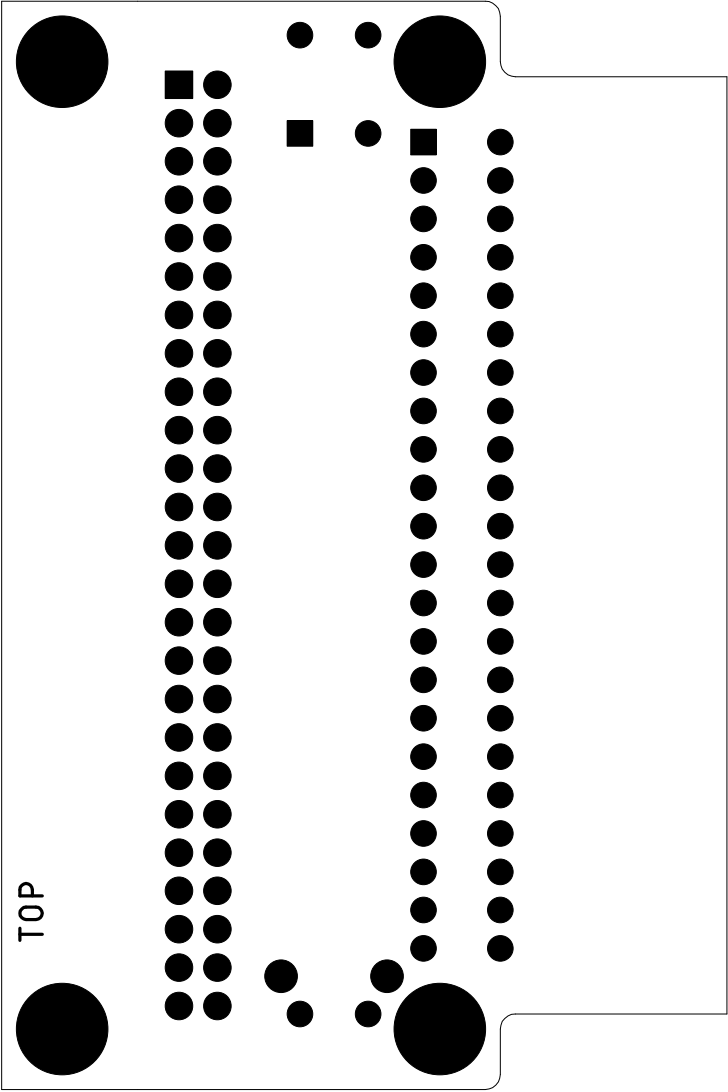
Sven Petersen 2026	Doc.-No.: 227-2-01-00	
	Cu: 35µm	Cu-Layers: 2
C64_Exp_cartridge		
22.05.2026 15:39		Rev.: 0
bottom		

Preliminary



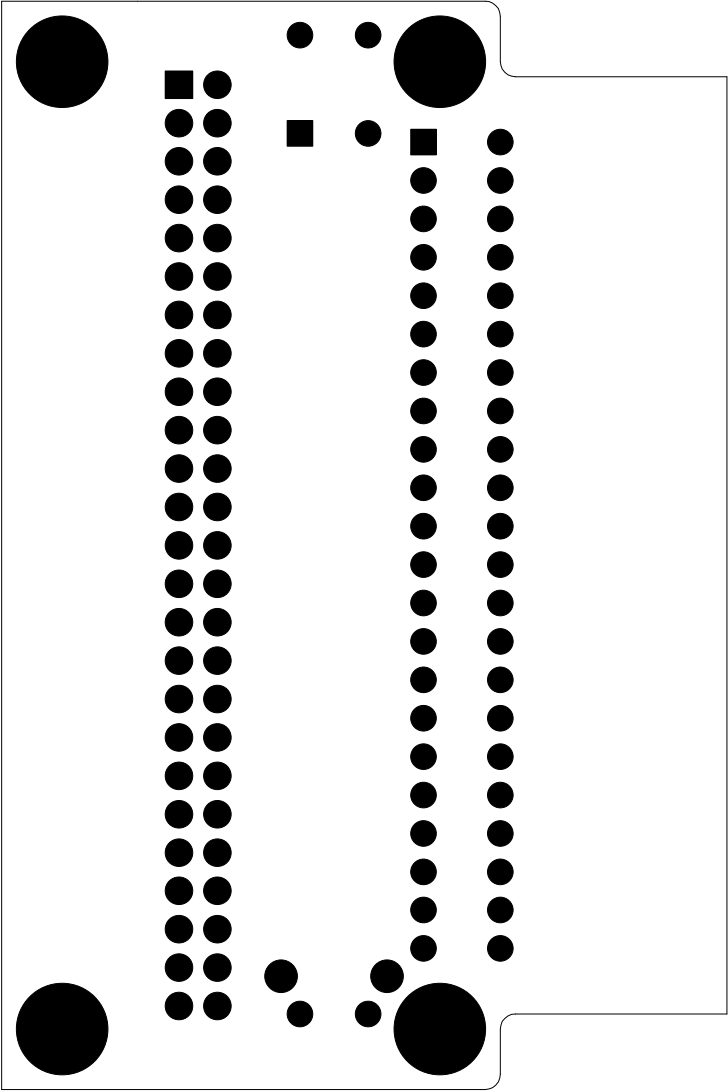
Sven Petersen 2026	Doc.-No.: 227-2-01-00	
	Cu: 35µm	Cu-Layers: 2
C64_Exp_cartridge		
22.05.2026 15:39		Rev.: 0
stopmask component side		

Preliminary



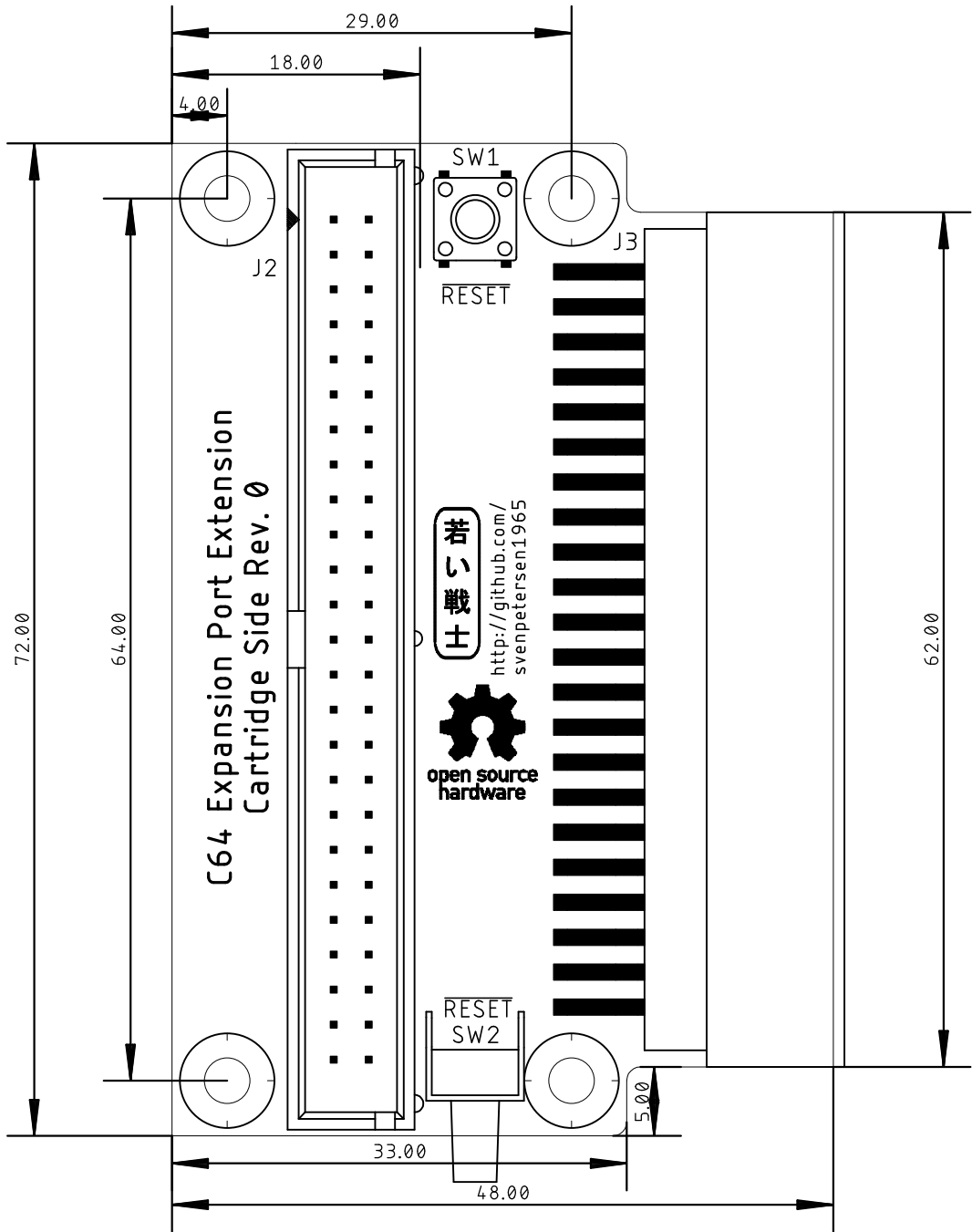
Sven Petersen 2026	Doc.-No.: 227-2-01-00	
	Cu: 35µm	Cu-Layers: 2
C64_Exp_cartridge		
22.05.2026 15:39		Rev.: 0
stopmask solder side		

Preliminary



Sven Petersen 2026	Doc.-No.: 227-2-01-00	
	Cu: 35μm	Cu-Layers: 2
C64_Exp_cartridge		
22.05.2026 15:39		Rev.: 0
placement component side		measures

Preliminary



C64 Expansion Port Extension Cartridge Side Rev. 0

Bill of Material Rev. 0.0

Pos.	Qty	Value	Footprint	Ref.-No.	Comment
1	1	227-2-01-00	2 Layer	PCB Rev. 0	2 layer, Cu 35μ, HASL, 72.0mm x 48.0mm, 1.6mm FR4
2	1	2x20 boxed header	2X25WV	J2	2x25, boxed pin header, e.g. reichelt.de WSL 40G
3	1	2x22 pin edge connector, 90°	EXPANSION_OU J3 T		1/5 x 5pc Industrial Card Edge Slot Socket Connector 22x2P 44P 2.54mm 0.1" Right Angle
4	1	JPT1236HB	JTP_1236HB	SW2	90° alternative reset switch, AliExpress search term "tact switch 90°"
5	1	JTP-1130	JTP-1130	SW1	Standard 6x6mm tact switch, e.g. Namae JTP-1130 or any other 6x6 tact switch, e.g. reichelt.de JTP-1130 or AliExpress search term "tact switch"